

Note: I have repeated few topics from last lecture.

Eigenvalues, eigenvectors of a square matrix

Let A be an $n \times n$ matrix. A number $\lambda \in \mathbb{R}$ is called an eigenvalue if there is a nonzero vector $v \in \mathbb{R}^n$, called an eigenvector,

$$\text{such that } Av = \lambda v.$$

Proposition. Let A be an $n \times n$ matrix. Then A has an eigenvector if and only if there is a line L in $\mathbb{R}^n$ such that $AL = L$.

Proof Let $v \neq 0$ be an eigenvector of A for an eigenvalue λ . Then, $Av = \lambda v$ and therefore

$$A(\alpha v) = \alpha Av = \alpha \lambda v = \lambda \alpha v.$$

This implies that if $L = \{\alpha v : \alpha \in F\}$ then

$$\begin{aligned} AL &= \{\alpha Av : \alpha \in F\} = \{\lambda \alpha v : \alpha \in F\} \\ &= \{\beta v : \beta \in F\} = L. \end{aligned}$$

Conversely, if $AL = L$ for some line L , then pick a nonzero vector $v \in L$ and observe that

$$L = \{ \lambda v : \lambda \in F \}.$$

Now $AL = L \Rightarrow Av \in L$. That is, $Av = \lambda v$ for some $\lambda \in F$. Thus, v is an eigenvector of A .

Proposition Let A be an $n \times n$ matrix. Then, $\det(A) = 0$ if and only if there is a nonzero element $v \in F^n$ such that $Av = 0$.

Proof

Let B be an invertible $n \times n$ matrix and $0 \neq v \in F^n$.

Then, $Bv \neq 0$ (why? $Bv = 0 \Rightarrow B^{-1}Bv = B^{-1}0 \Rightarrow v = 0$)

$(\Rightarrow)$ Let $\det A = 0$. Then $\det(A^T) = 0$. There are elementary matrices $B_1, B_2, \dots, B_\ell$ such that

$B_1 B_2 \cdots B_\ell A^T = J$, where J is row reduced echelon form of A^T . Note that the last row of J is a Zero row. Now,

$$A B_\ell^T B_{\ell-1}^T \cdots B_2^T B_1^T = J^T.$$

Observe that

- Since each B_i is invertible, so is each B_i^T .
- $J^T e_n = 0$.

Now, $B_1^T e_n \neq 0 \Rightarrow B_2^T B_1^T e_n \neq 0 \Rightarrow \dots \Rightarrow \underbrace{B_\ell^T \dots B_2^T B_1^T}_{\mathcal{V}} e_n \neq 0$

Thus $A \mathcal{V} = A B_\ell^T B_{\ell-1}^T \dots B_2^T B_1^T e_n = J^T e_n = 0$; where $\mathcal{V} \neq 0$

$\Leftarrow$ Let $\ker(A) \neq \{0\}$, that is, there is $0 \neq \mathcal{V} \in \mathbb{F}^n$ such that $A \mathcal{V} = 0$. If $\det(A) \neq 0$ then A is invertible and we obtain $\mathcal{V} = A^{-1} A \mathcal{V} = A^{-1} 0 = 0$, a contradiction.
Thus $\det(A) = 0$

Method of finding eigenvalues & eigenvectors of an $n \times n$ matrix

Let A be an $n \times n$ matrix. We need to find all $\lambda \in \mathbb{R}$ so that $A \mathcal{V} = \lambda \mathcal{V}$ for some $\mathcal{V} \neq 0$.

That is, find all $\mathcal{V} \neq 0$ so that $(A - \lambda I_n) \mathcal{V} = A \mathcal{V} - \lambda I_n \mathcal{V} = A \mathcal{V} - \lambda \mathcal{V} = 0$

Thus we need to calculate $\ker(A - \lambda I_n)$.

Note that $A - \lambda I_n$ is a $n \times n$ matrix and therefore there is

a nonzero $\mathbf{v}$ such that $(A - \lambda I_n)\mathbf{v} = 0$ if and only if $\det(A - \lambda I_n) = 0$

Steps

- (i) Calculate $\det(A - \lambda I_n) = 0$. You will get a polynomial in λ .
- (ii) Find all roots of the polynomial in $\mathbb{R}$, if it exist. Those are the eigenvalues.
- (iii) For those eigenvalues λ , $\ker(A - \lambda I_n) \neq \{0\}$.
Thus any $0 \neq \mathbf{v} \in \ker(A - \lambda I_n)$ is an eigenvector for that eigenvalue λ .

Example / non examples

- (1) Find all eigenvectors and eigenvalues of $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = A$.

We need to find all $\lambda \in \mathbb{R}$ such that $A\mathbf{v} = \lambda\mathbf{v}$.

$$\begin{aligned} \text{(i) Now } \det(A - \lambda I_2) &= \det\left(\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right) \\ &= \det\begin{pmatrix} 1-\lambda & 2 \\ 0 & 3-\lambda \end{pmatrix} \end{aligned}$$

$$= (1-\lambda)(3-\lambda)$$

Then, $\det(A - \lambda I_2) = 0$ for $\lambda = 1$ and $\lambda = 3$.

(ii) Eigenvalues: $\lambda = 1, \lambda = 3$.

(iii) Eigenvectors: Let $\lambda = 1$. Then $A - \lambda I_2 = \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix}$.

Any nonzero vector $\begin{pmatrix} x \\ y \end{pmatrix} \in \text{Ker}(A - \lambda I_2)$ is an eigenvector.

$$\text{Since, } \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow 2y = 0 \Rightarrow y = 0.$$

We see that any vector of the form $\begin{pmatrix} x \\ 0 \end{pmatrix}$, where $x \neq 0$, is an eigenvector.

In particular, $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \text{Ker} \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix}$ is an

eigenvector of A for the eigenvalue $\lambda = 1$.

Let $\lambda = 3$

Then, we need to find the kernel of

$$\begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix}.$$

Let $\begin{pmatrix} x \\ y \end{pmatrix} \in \text{Ker} \begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix}$. Then $\begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\Rightarrow -2x + 2y = 0 \Rightarrow x = y.$$

Thus $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \in \mathbb{R}^2$ is an eigenvector of A for eigenvalue $\lambda=3$.

② Consider the matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{aligned} \text{Then } \det(A - \lambda I_2) &= \det \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} \\ &= \lambda^2 + 1. \end{aligned}$$

Since $\lambda^2 + 1 = 0$ has no roots in $\mathbb{R}$,
there are no eigenvalues in $\mathbb{R}$ and therefore no eigenvectors in $\mathbb{R}^2$.

Eigenvector Basis

We consider example ①.

$$A: \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \quad A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$$

$$\text{Note that } A e_1 = 1 \cdot e_1 + 0 \cdot e_2$$

$$A e_2 = 2 e_1 + 3 e_2$$

So the matrix of linear transformation is A , with respect to the basis $\{e_1, e_2\}$.

We know that $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are eigenvectors for

Eigenvalues $\lambda=1$ and $\lambda=3$ respectively. Also observe that $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ is a basis of $\mathbb{R}^2$.

Now,

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Therefore, the matrix of the linear transformation A with respect to the basis $\mathcal{B}$ is

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Interpretation of M

$\begin{pmatrix} x \\ y \end{pmatrix}$ is mapped to $\begin{pmatrix} x+2y \\ 3y \end{pmatrix}$ by A

We have $A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+2y \\ 3y \end{pmatrix}.$

Write

$$\begin{pmatrix} x \\ y \end{pmatrix} = (x-y) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Consider the equation

$$\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x-y \\ y \end{pmatrix} = \begin{pmatrix} x-y \\ 3y \end{pmatrix}$$

We interpret this equation Correctly and make sense as follows

$$\begin{aligned}\begin{pmatrix} x \\ y \end{pmatrix} &= (x-y)\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ is mapped to } (x-y)\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3y\begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} x-y \\ 0 \end{pmatrix} + \begin{pmatrix} 3y \\ 3y \end{pmatrix} \\ &= \begin{pmatrix} x+2y \\ 3y \end{pmatrix}\end{aligned}$$

by A.

Similarly,

$$\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ 3y \end{pmatrix}$$

Interpreting $\begin{pmatrix} x \\ y \end{pmatrix}$ as $x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and

$\begin{pmatrix} x \\ 3y \end{pmatrix}$ as $x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3y\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, we obtain that

$\begin{pmatrix} x+y \\ y \end{pmatrix}$ is mapped to $\begin{pmatrix} x+3y \\ 3y \end{pmatrix}$ by A (Check that this is correct!)